Project Proposal — 2026
Embedded Real-Time Systems
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# Autonomous Pharmacy Robot for Real-Time Medication Preparation

*Simulation of a hospital pharmacy using ROS2, TurtleBot3 and Gazebo — with priority-based real-time scheduling and deadline enforcement*

| Hard RT | Soft RT | Firm RT | Priority Scheduling | Watchdog | Recovery | Nav2 | TurtleBot3 | Gazebo |
|---|---|---|---|---|---|---|---|---|


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**Team Name**
**Team PharmaBot**
**Module**
Embedded Real-Time Systems
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## 1. Main Objective

The objective of this project is to design and simulate an **autonomous pharmacy robot** using ROS2 and TurtleBot3, capable of preparing and delivering medications according to the urgency of patient requests. The system simulates a hospital pharmacy where different medication requests arrive with different urgency levels, and the robot must process them under strict real-time constraints.

In real-time systems, correctness depends not only on the logical result but also on **the time at which the result is produced**. Respecting deadlines is therefore a key aspect of this project. The robot uses **priority-based scheduling** to handle three types of requests:

| Priority Level | Deadline Type | Example Request | If deadline is missed |
|---|---|---|---|
| **CRITICAL — Reanimation** | Hard RT | Emergency medication | Fatal — alert + recovery mode |
| **URGENT — Emergency ward** | Soft RT | Post-op pain management | Delay tolerated, quality degraded |

| | | | |
|---|---|---|---|
| **STANDARD — Consultation** | Firm RT | Routine prescription | Cancelled if deadline passed |

The robot will navigate inside a simulated pharmacy environment using ROS2, Gazebo and Nav2, retrieve medications from storage locations and deliver them to the correct service. This project directly illustrates the **classification of real-time systems (hard, soft, and firm deadlines)** studied in the course.

## 2. Expected Outcomes

At the end of this project, we expect to deliver:

- **A ROS2-based autonomous robotic system** capable of navigating in a simulated pharmacy environment (Gazebo + RViz2).
- **An implementation of real-time scheduling mechanisms** to manage medication requests with hard, soft and firm deadlines.
- **A live dashboard** (matplotlib) showing the medication queue, countdown timers, and robot status in real time.
- **A watchdog and automatic recovery node** that detects missed hard deadlines and triggers a safe fallback mode.
- **A GitHub repository** containing the complete project code, documentation and setup instructions.
- **A demonstration** illustrating how real-time constraints affect task execution and system behavior under normal and overload conditions.

The project will demonstrate how real-time concepts such as **deadlines, scheduling and task priorities** can be applied to a realistic healthcare scenario — directly connecting to the infamous failures studied in class (**Apollo 11, Ariane 5**).

## 3. Technical Stack

| Component | Technology | Role in the project |
|---|---|---|
| **Robot & Simulation** | TurtleBot3 + Gazebo | Physical robot simulation and navigation environment |
| **Navigation** | ROS2 Nav2 + RViz2 | Autonomous path planning, obstacle avoidance, live map visualisation |
| **RT Scheduler** | Python Priority Queue | Dynamic task ordering based on deadline type and urgency level |
| **Watchdog/Recovery** | Custom ROS2 node | Detects missed hard deadlines and triggers safe fallback mode |
| **Live Dashboard** | matplotlib + rclpy | Real-time queue with countdown timers and robot status |

### Connection to the course

This project directly applies the core concepts of the module: **hard, soft and firm deadline classification**; **priority-based scheduling** and task preemption; **temporal constraints on I/O tasks**; and **system predictability** (the 'no surprises' principle). The watchdog mechanism also illustrates **fault tolerance** in safety-critical systems — directly connecting to the infamous failures studied in class (**Apollo 11, Ariane 5**).